

Orchestration for the Edge-Cloud Continuum

What is the Edge-Cloud Continuum?

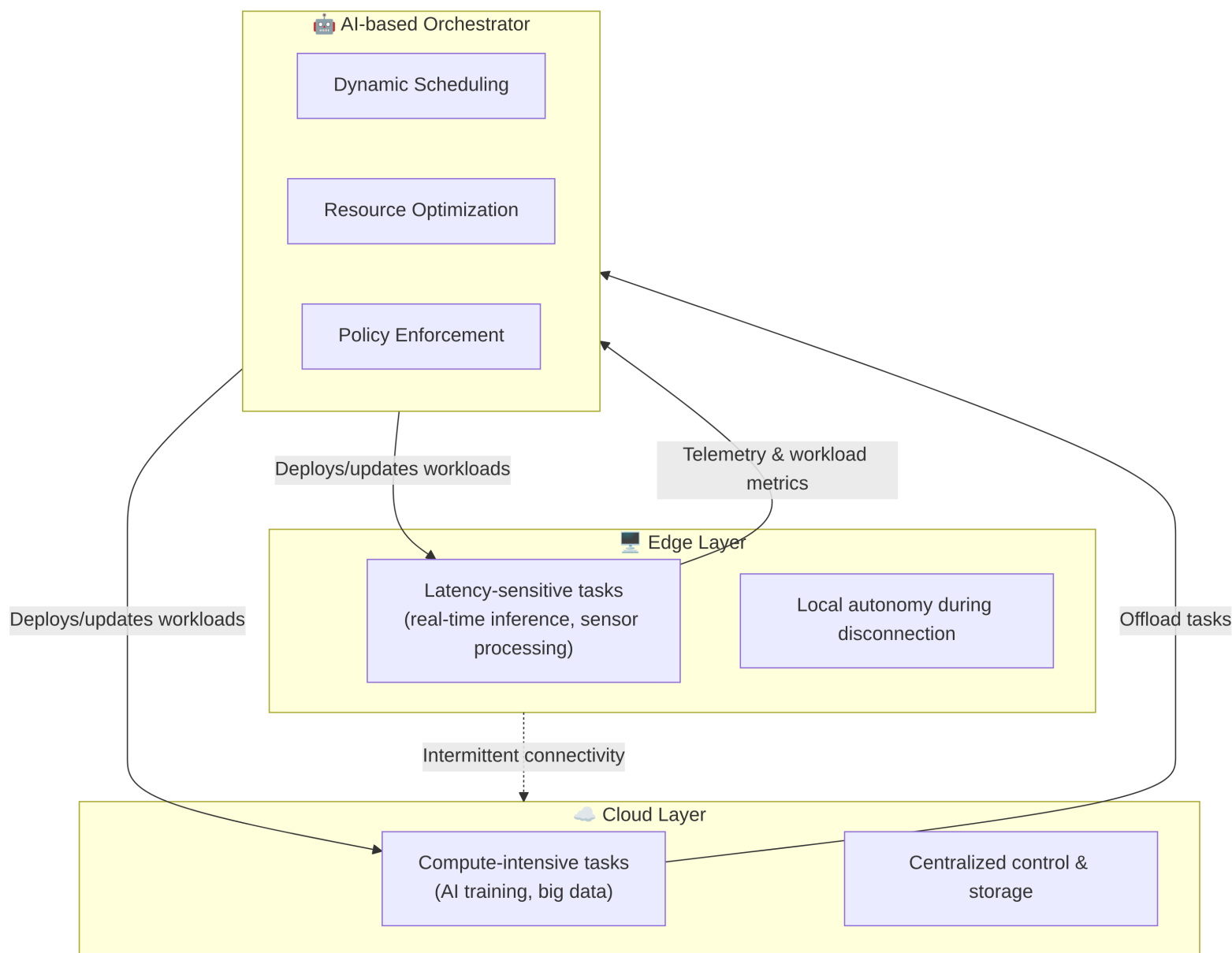
- **Cloud:** centralized, high-performance infrastructure (data centers)
 - **Edge:** computing resources closer to users/devices
 - **Continuum:** seamless collaboration between cloud and edge resources
 - **Goal:** deploy, manage, and scale workloads dynamically across both environments
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Why Orchestration?

- Edge and cloud environments are **heterogeneous**
 - Applications are **distributed**, often involving:
 - IoT devices
 - Real-time analytics
 - AI inference
 - Need for **automation** in:
 - Deployment
 - Scaling
 - Resilience
 - Monitoring
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Toward Unified and Adaptive Orchestration

- **Manages both latency-sensitive and compute-intensive tasks**
 - Deploys **real-time workloads** (e.g., sensor fusion, anomaly detection) close to data sources at the edge.
 - Offloads **compute-heavy operations** (e.g., model training, batch analytics) to the cloud.
 - Continuously adapts based on **network latency**, **resource availability**, and **QoS requirements**.
- **Enables autonomous operation during disconnection**
 - Edge nodes maintain **local decision-making** even when cloud connectivity is lost.
 - Synchronization and reconciliation occur automatically when the connection is restored.
 - Ensures **service continuity** and **resilience** in environments with intermittent networks.
- **Support scheduling and resource optimization**
 - Uses **machine learning models** to predict workload demands and adjust resource allocation.
 - Dynamically balances **power efficiency**, **performance**, and **cost** across the continuum.
 - Learns from telemetry data (CPU, GPU, network) to refine orchestration decisions over time.



Kubernetes (K8s)

- **De facto standard** for container orchestration in the cloud
- Provides:
 - Automated deployment and scaling
 - Service discovery and load balancing
 - Self-healing (restarts, rescheduling)
- Suitable for **large-scale** clusters
- Challenges at the edge:
 - High overhead
 - Limited connectivity and resources

KubeEdge: Extending K8s to the Edge

- Open-source project from **CNCF**
- Extends Kubernetes **beyond the cloud** to edge nodes
- Architecture:
 - **CloudCore:** runs in the cloud (control plane)
 - **EdgeCore:** runs on edge devices (data plane)

- Features:
 - Offline tolerance
 - Device management APIs
 - Synchronization between edge and cloud
 - Works with AI and IoT workloads
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K3s: Lightweight Kubernetes

- **Optimized for edge and IoT** environments
 - Developed by Rancher (now SUSE)
 - Key features:
 - Single binary (<100 MB)
 - Minimal dependencies
 - SQLite instead of etcd (by default)
 - ARM-compatible
 - Ideal for **resource-constrained devices**
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MicroK8s: Developer-Friendly Kubernetes

- Developed by **Canonical** (Ubuntu)
- **Zero-ops** Kubernetes for:
 - Local development
 - Edge and IoT
- Key features:
 - Easy installation: `snap install microk8s --classic`
 - Modular via `microk8s enable <addon>`
 - Supports clustering (HA setups)
- Runs on Linux, macOS, and Windows

References

- CNCF KubeEdge: <https://kubedge.io>
- SUSE K3s: <https://k3s.io>
- Canonical MicroK8s: <https://microk8s.io>
- Kubernetes Docs: <https://kubernetes.io>